assignment_6_solutions_rubric

November 20, 2023

0.1 Problem 1a: Install COSMIC and run the first example; then print the bpp with the tphys, mass_1, mass_2, kstar_1, kstar_2, porb, ecc, and evol_type columns

6 points for installation (guaranteed if the rest of the assignment is done)

3 points for running the binary and printing

3 points for words description

[1]:
```
from cosmic.sample.initialbinarytable import InitialBinaryTable
from cosmic.evolve import Evolve
import matplotlib.pyplot as plt
import numpy as np

from cosmic.plotting import evolve_and_plot
```

[2]:
```
single_binary = InitialBinaryTable.InitialBinaries(m1=85.543645, m2=84.99784,␣
↪porb=446.795757, ecc=0.448872, tphysf=13700.0, kstar1=1, kstar2=1,␣
↪metallicity=0.002)
```

[3]:
```
BSEDict = {'xi': 1.0, 'bhflag': 1, 'neta': 0.5, 'windflag': 3, 'wdflag': 1,␣
↪'alpha1': 1.0, 'pts1': 0.001, 'pts3': 0.02, 'pts2': 0.01, 'epsnov': 0.001,␣
↪'hewind': 0.5, 'ck': 1000, 'bwind': 0.0, 'lambdaf': 0.0, 'mxns': 3.0, 'beta':
↪ -1.0, 'tflag': 1, 'acc2': 1.5, 'grflag' : 1, 'remnantflag': 4, 'ceflag': 0,␣
↪'eddfac': 1.0, 'ifflag': 0, 'bconst': 3000, 'sigma': 265.0, 'gamma': -2.0,␣
↪'pisn': 45.0, 'natal_kick_array' : [[-100.0,-100.0,-100.0,-100.0,0.0], [-100.
↪0,-100.0,-100.0,-100.0,0.0]], 'bhsigmafrac' : 1.0, 'polar_kick_angle' : 90,␣
↪'qcrit_array' : [0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.
↪0,0.0], 'cekickflag' : 2, 'cehestarflag' : 0, 'cemergeflag' : 0, 'ecsn' : 2.
↪25, 'ecsn_mlow' : 1.6, 'aic' : 1, 'ussn' : 0, 'sigmadiv' :-20.0, 'qcflag' :␣
↪1, 'eddlimflag' : 0, 'fprimc_array' : [2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.
↪0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.
↪0,2.0/21.0,2.0/21.0,2.0/21.0], 'bhspinflag' : 0, 'bhspinmag' : 0.0,␣
↪'rejuv_fac' : 1.0, 'rejuvflag' : 0, 'htpmb' : 1, 'ST_cr' : 1, 'ST_tide' : 1,␣
↪'bdecayfac' : 1, 'rembar_massloss' : 0.5, 'kickflag' : 0, 'zsun' : 0.014,␣
↪'bhms_coll_flag' : 0, 'don_lim' : -1, 'acc_lim' : -1, 'rtmsflag' : 0}
```

```
[4]: bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,
     ↪BSEDict=BSEDict)
```

```
[5]: print(bpp[['tphys', 'mass_1', 'mass_2', 'kstar_1', 'kstar_2', 'porb', 'ecc',
     ↪'evol_type']])
```

```
          tphys     mass_1      mass_2  kstar_1  kstar_2        porb  \
0      0.000000  85.543645   84.997840      1.0      1.0  446.795757
0      3.717075  72.720332   72.376321      2.0      1.0  617.238959
0      3.718373  72.589218   72.377845      2.0      1.0  252.923207
0      3.720012  71.959504   72.806393      4.0      1.0  252.533894
0      3.740038  32.352601  110.573279      4.0      1.0  628.743501
0      3.741215  32.176152  110.618802      7.0      1.0  628.922011
0      4.071374  25.488585  106.891756      7.0      1.0  731.786530
0      4.071374  24.988585  106.891756     14.0      1.0  737.361815
0      4.894369  24.988590   88.885767     14.0      2.0  988.424464
0      4.895887  24.989628   88.681512     14.0      2.0  986.044004
0      4.896841  24.990676   88.469775     14.0      4.0  986.208557
0      4.896841  24.990676   88.469775     14.0      4.0  986.208557
0      4.896841  24.990676   41.981621     14.0      7.0   81.319009
0      4.896841  24.990676   41.981621     14.0      7.0   81.319009
0      5.205274  24.990687   31.554344     14.0      7.0  114.082486
0      5.205274  24.990687   31.054344     14.0     14.0  116.140982
0  13700.000000  24.990687   31.054344     14.0     14.0  116.033619

        ecc  evol_type
0  0.448872        1.0
0  0.448872        2.0
0  0.000000        3.0
0  0.000000        2.0
0  0.000000        4.0
0  0.000000        2.0
0  0.000000       15.0
0  0.003791        2.0
0  0.003791        2.0
0  0.000000        3.0
0  0.000000        2.0
0  0.000000        7.0
0  0.000000        8.0
0  0.000000        4.0
0  0.000000       16.0
0  0.008921        2.0
0  0.008921       10.0
```

0.2 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 3.7 Myr. The primary fills its Roche lobe while it is on the Hertzsprung Gap and the mass transfer remains stable transferring over 40 solar masses of material to the companion. This results in the binary having a wider orbit than it started out with. Eventually, the primary envelope is fully stripped, then the primary forms a black hole with 25 solar masses. The formation of the black hole doesn't change the orbit very much, so the natal kick is probably small. The secondary then evolves off the main sequence at 4.9 Myr and fills its Roche lobe on the Hertzsprung Gap. Once the secondary evolves to core helium burning, the mass transfer becomes unstable and leads to a common envelope evolution which produces a binary with an 81 day orbital period. The black hole formed by the primary doesn't accrete much material. The secondary forms a 31 solar mass black hole and the orbital period remains at 116 days even at a Hubble time of evolution.

0.5 pt off for missing Roche lobe overflow statement (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes

0.5 pt off for not describing how the masses change

0.3 Problem 1b: Evolve a ZAMS binary with

0.3.1 $mass\_1 = 6.0\,\mathrm{M}_{\odot}$, $mass\_2 = 1.5\,\mathrm{M}_{\odot}$, $porb = 600\,\mathrm{days}$, $ecc = 0$, $Z = 0.014$

3 points for running the binary and printing

3 points for orbital period plot

3 points for mass plot

3 points for words description

[6]:
```
single_binary = InitialBinaryTable.InitialBinaries(m1=6.0, m2=1.5, porb=600.0,␣
↪ecc=0.0, tphysf=13700.0,
                                                    kstar1=1, kstar2=1,␣
↪metallicity=0.014)
bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,␣
↪BSEDict=BSEDict, dtp=0.0)
bpp[['tphys', 'kstar_1', 'kstar_2', 'mass_1', 'mass_2', 'porb', 'evol_type']]
```

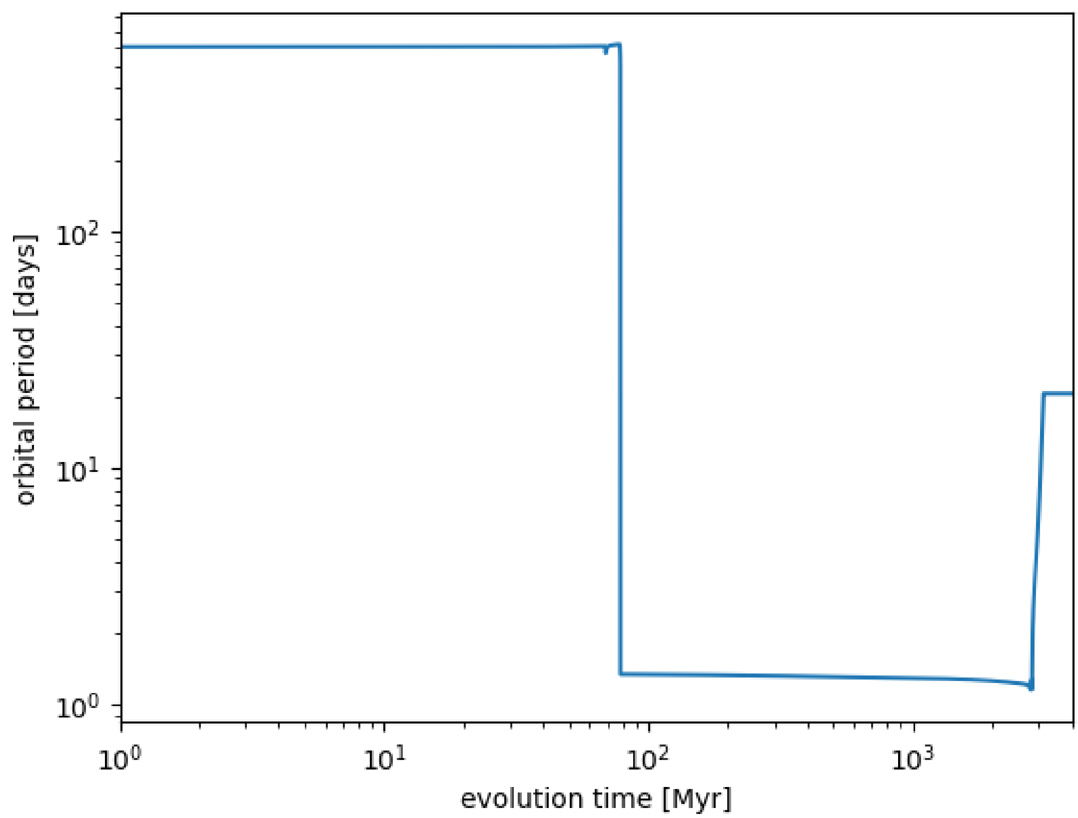
[6]:

	tphys	kstar_1	kstar_2	mass_1	mass_2	porb	evol_type
0	0.000000	1.0	1.0	6.000000	1.500000	600.000000	1.0
0	68.559582	2.0	1.0	5.979993	1.500000	603.213995	2.0
0	68.809728	3.0	1.0	5.979524	1.500001	603.349846	2.0
0	68.918809	4.0	1.0	5.978294	1.500014	578.837101	2.0
0	78.038656	5.0	1.0	5.890052	1.500761	600.803163	2.0
0	78.347085	5.0	1.0	5.875955	1.501061	489.380684	3.0
0	78.347085	5.0	1.0	5.875955	1.501061	489.380684	7.0
0	78.347085	8.0	1.0	1.493853	1.501061	2.964431	8.0
0	78.347085	8.0	1.0	1.493853	1.501061	2.964431	7.0
0	78.347085	11.0	1.0	0.932782	1.501061	1.339521	8.0
0	78.347085	11.0	1.0	0.932782	1.501061	1.339521	4.0

0	2723.537548	11.0	2.0	0.932782	1.501061	1.201748	2.0
0	2725.324049	11.0	2.0	0.932783	1.501055	1.205693	3.0
0	2868.091958	11.0	3.0	0.933814	0.560470	2.729554	2.0
0	3129.365737	11.0	3.0	0.942573	0.227356	20.542647	4.0
0	3139.028907	11.0	10.0	0.942784	0.226925	20.546670	2.0
0	13700.000000	11.0	10.0	0.942784	0.226925	20.546565	10.0

[7]:
```
plt.plot(bcm.tphys, bcm.porb)
plt.xlim(1, 4000)
plt.xscale('log')
plt.yscale('log')
plt.ylabel('orbital period [days]')
plt.xlabel('evolution time [Myr]')
```

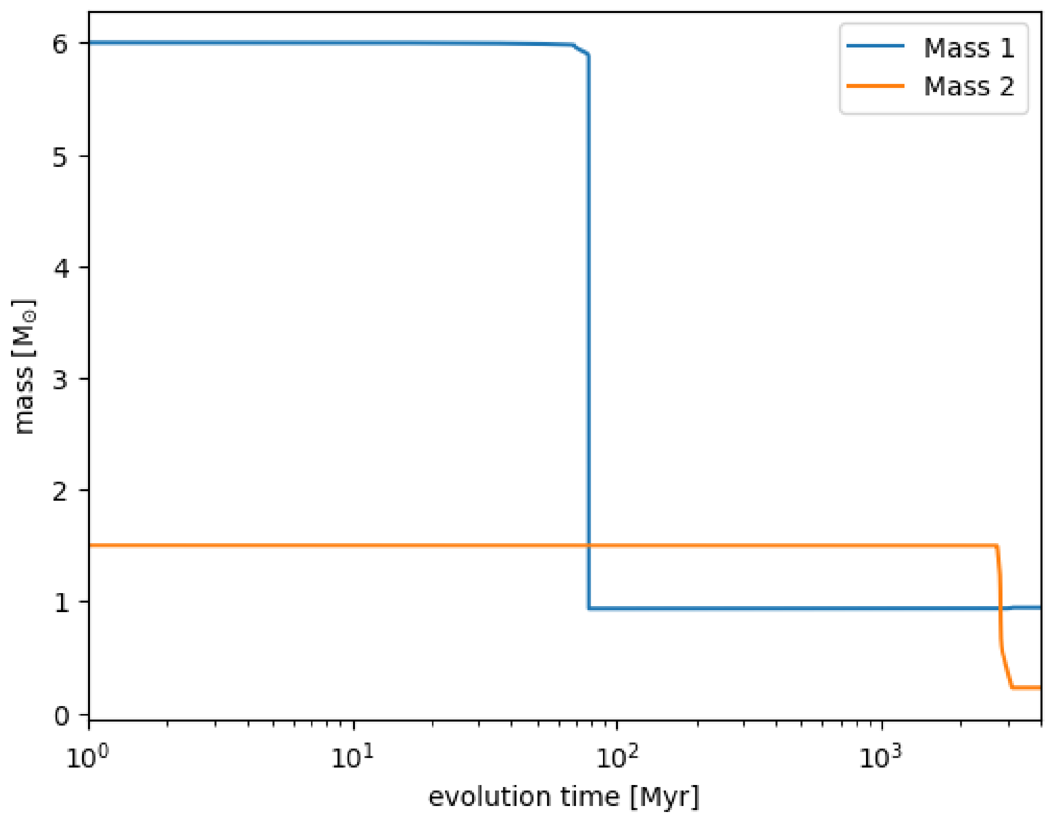
[7]: Text(0.5, 0, 'evolution time [Myr]')



[8]:
```
plt.plot(bcm.tphys, bcm.mass_1, label='Mass 1')
plt.plot(bcm.tphys, bcm.mass_2, label='Mass 2')
plt.xlim(1, 4000)
plt.xscale('log')
plt.legend()
```

```
plt.xlabel('evolution time [Myr]')
plt.ylabel('mass [M$_{\odot}$]')
```

[8]: Text(0, 0.5, 'mass [M$_{\\odot}$]')



[ ]:

0.4 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 68.6 Myr and subsequently evolves across the Hertsprung Gap, climbs the red giant branch, and begins core helium burning. The primary fills its Roche lobe while it is on the early asymptotic giant branch and the mass transfer becomes unstable. This results in a common envelope evolution that leaves behind a much shorter orbit. Since the Roche overflow resulted in a common envelope, the mass of the secondary didn't increase at all, but the mass of the primary was significantly decreased. At ~2.6 Gyr the secondary evolves off the main sequence and fills it's Roche lobe while it is crossing the Hertzsprung Gap. The mass transfer remains stable, but the WD can't accrete much material, so there is little mass exchange and the secondary just loses mass. There is a kink in the time evolution of the secondary because the secondary starts to ascend the giant branch. The orbital period initially decreases, but then increases because there was a mass ratio inversion and the secondary mass changes from being more massive than the primary to being less massive than the primary. The end stage is a Carbon-Oxygen and Helium white dwarf binary with an orbital period of 20 days.

0.5 pt off for missing Roche lobe overflow statement going to CE for primary and Stable for secondary (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes

0.5 pt off for not describing how the masses change

0.5 Problem 1c: Evolve a ZAMS binary with a random seed of 5 and

0.5.1 $mass\_1 = 50.0\,\mathrm{M}_{\odot}$, $mass\_2 = 35\,\mathrm{M}_{\odot}$, $porb = 120\,\mathrm{days}$, $ecc = 0$, $Z = 0.014$

3 points for running the binary and printing

3 points for orbital period plot

3 points for mass plot

3 points for words description

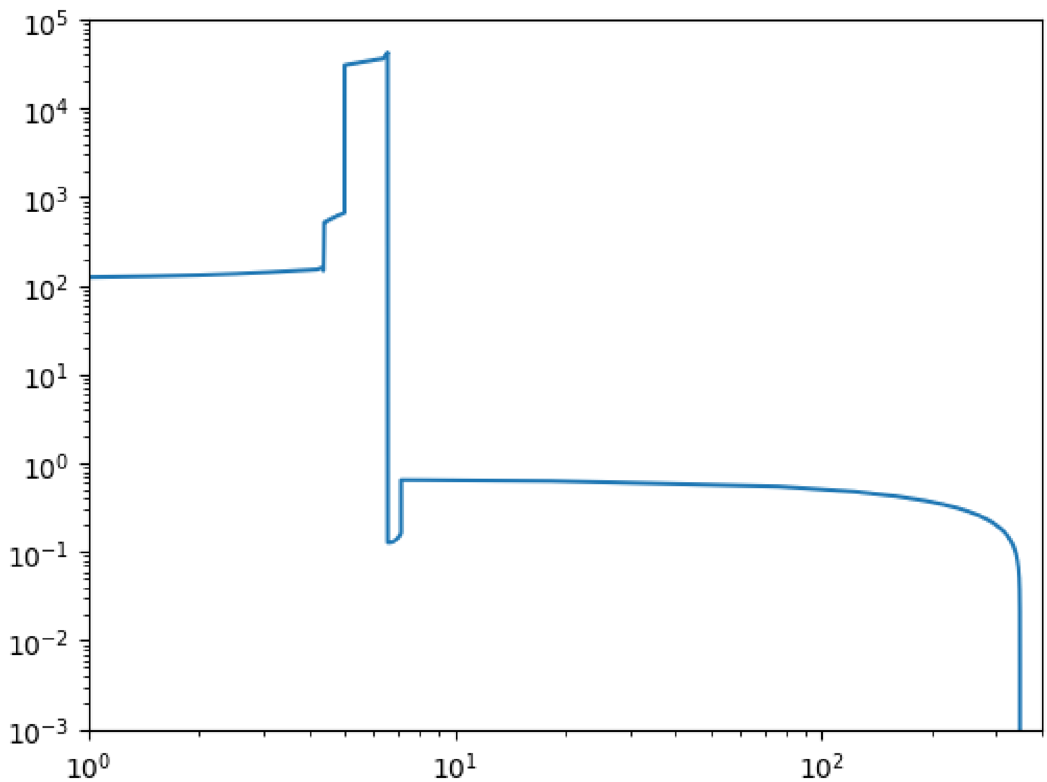
[9]:
```
np.random.seed(5)
single_binary = InitialBinaryTable.InitialBinaries(m1=50.0, m2=35.0, porb=120.
 ↪0, ecc=0.0, tphysf=13700.0,
                                                        kstar1=1, kstar2=1,␣
 ↪metallicity=0.014)
bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,␣
 ↪BSEDict=BSEDict, dtp=0.0)
print(bpp[['tphys', 'mass_1', 'mass_2', 'kstar_1', 'kstar_2', 'porb', 'ecc',␣
 ↪'evol_type']].round(3))
```

```
      tphys  mass_1  mass_2  kstar_1  kstar_2       porb    ecc  evol_type
0     0.000  50.000  35.000      1.0      1.0    120.000  0.000        1.0
0     4.358  40.464  32.495      2.0      1.0    162.882  0.000        2.0
0     4.361  40.436  32.495      2.0      1.0    162.923  0.000        3.0
0     4.364  40.032  32.854      4.0      1.0    161.846  0.000        2.0
0     4.392  14.802  57.774      4.0      1.0    521.423  0.000        4.0
0     4.397  14.736  57.803      7.0      1.0    520.661  0.000        2.0
0     4.954   8.846  55.665      8.0      1.0    658.361  0.000        2.0
0     4.984   8.554  55.544      8.0      1.0    666.877  0.000       15.0
0     4.984   4.180  55.544     14.0      1.0  30894.690  0.921        2.0
0     6.538   4.180  46.884     14.0      2.0  42227.385  0.921        2.0
0     6.542   4.180  46.486     14.0      2.0    956.646  0.000        3.0
0     6.542   4.180  46.486     14.0      2.0    956.646  0.000        7.0
0     6.542   4.180  17.623     14.0      7.0      0.158  0.000        8.0
0     6.542   4.180  17.623     14.0      7.0      0.158  0.000        4.0
0     7.095   4.269   9.562     14.0      8.0      0.157  0.000        2.0
0     7.123   4.275   9.248     14.0      8.0      0.165  0.000       16.0
```

```
0      7.123   4.275   4.984    14.0    14.0     0.649  0.593      2.0
0    348.667   4.275   4.984    14.0    14.0     0.000  0.000      3.0
0    348.667   9.259   0.000    14.0    15.0     0.000 -1.000      6.0
0  13700.000   9.259   0.000    14.0    15.0     0.000 -1.000     10.0
```

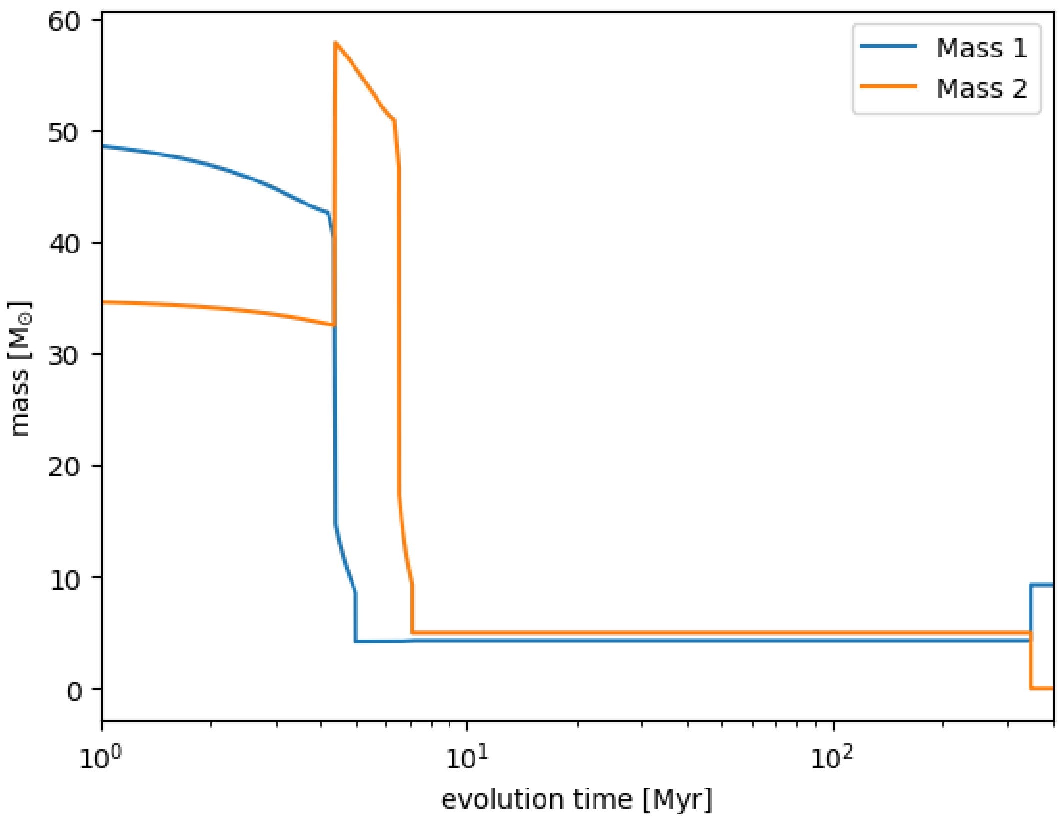
[10]:
```
plt.plot(bcm.tphys, bcm.porb)
plt.yscale('log')
plt.xscale('log')
plt.xlim(1, 400)
plt.ylim(1e-3, 1e5)
```

[10]:
```
(0.001, 100000.0)
```



[11]:
```
plt.plot(bcm.tphys, bcm.mass_1, label='Mass 1')
plt.plot(bcm.tphys, bcm.mass_2, label='Mass 2')
plt.xlim(1, 400)
plt.xscale('log')
plt.legend()
plt.xlabel('evolution time [Myr]')
plt.ylabel('mass [M$_{\odot}$]')
```

[11]:
```
Text(0, 0.5, 'mass [M$_{\\odot}$]')
```



[ ]:

[ ]:

0.6 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 4.358 Myr. The primary fills its Roche lobe while it is on the Hertzsprung Gap and the mass transfer remains stable transferring 35 solar masses of material to the companion. This results in the binary having a wider orbit than it started out with. Eventually, the primary envelope is fully stripped, and the left over helium star loses mass from stellar winds which further widens the orbit. The primary forms a black hole with 4 solar masses at 4.99 Myr. The formation of the black hole changes the orbit significantly by both widening and increasing the eccentricity, so the natal kick is probably strong. The secondary then evolves off the main sequence at 6.5 Myr and fills its Roche lobe on the Hertzsprung Gap and immediately leads to a common envelope evolution which produces a binary with an 0.16 day orbital period. The black hole formed by the primary doesn't accrete during the common envelope. The secondary forms a 4.98 solar mass black hole at 7.1 Myr and the orbital period increases to 0.35 days. The two black holes merge at 349 Myr. The main differences between this evolution and the evolution in 1a are the different black hole masses and the fact that the black holes in 1a don't merge in a Hubble time, but this black hole would merge and produce gravitational waves.

0.5 pt off for missing Roche lobe overflow statement (stable for primary and CE for secondary) (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes (especially due to kicks)

0.5 pt off for not describing how the masses change

0.5 pt off for not comparing to the evolution of 1a.

0.7 Problem 2, 6 pts total

[12]:
```
import astropy.units as u
import astropy.constants as c
```

0.8 In the class notes:

The strain is

$h^2(\mathcal{M}_c, f_{\rm GW}, D) = \frac{128}{5}\frac{(G\mathcal{M}_c)^{10/3}}{c^8 D^2}\left(\pi f_{\rm GW}\right)^{4/3}$

and the chirp is

$\dot{f}(\mathcal{M}_c, f_{\rm GW}) = \frac{96}{5}\frac{(G\mathcal{M}_c)^{5/3}}{c^5}(2\pi f_{\rm orb})^{11/3}$

where $f_{\rm GW} = 2\, f_{\rm orb}$.

We can solve for the chirp mass from $\dot{f}$ as

$\mathcal{M}_c = \left(\frac{96}{5}\frac{\dot{f}}{(2\pi f_{\rm orb})^{11/3}}\right)^{3/5}\frac{c^3}{G}$

and for the distance as

$D = \sqrt{\frac{128}{5}\frac{(G\mathcal{M}_c)^{10/3}}{c^8 h^2}\left(\pi f_{\rm GW}\right)^{4/3}}$

0.9 The class notes are missing a π in the denominator of the chirp equation!:

The strain is

$h^2(\mathcal{M}_c, f_{\rm GW}, D) = \frac{128}{5}\frac{(G\mathcal{M}_c)^{10/3}}{c^8 D^2}\left(\pi f_{\rm GW}\right)^{4/3}$

and the chirp is

$\dot{f}(\mathcal{M}_c, f_{\rm GW}) = \frac{96}{5\pi}\frac{(G\mathcal{M}_c)^{5/3}}{c^5}(2\pi f_{\rm orb})^{11/3}$

where $f_{\rm GW} = 2\, f_{\rm orb}$.

We can solve for the chirp mass from $\dot{f}$ as

$\mathcal{M}_c = \left(\frac{96}{5}\frac{\dot{f}}{(2\pi f_{\rm orb})^{11/3}}\right)^{3/5}\frac{c^3}{G}$

and for the distance as

$D = \sqrt{\frac{128}{5}\frac{(G\mathcal{M}_c)^{10/3}}{c^8 h^2}\left(\pi f_{\rm GW}\right)^{4/3}}$

```
[13]: def chirp_mass_from_chirp_from_class(f_dot, f_gw):
          return 1/c.G * ((5/96) *c.c**5 * f_dot * (np.pi * f_gw)**(-11/3))**(3/5)

      def distance_from_strain_chirp_mass(strain, chirp_mass, f_gw):
          dist = np.sqrt(128/5 / strain**2 * (np.pi * f_gw)**(4/3) * (c.G *
       ↪chirp_mass)**(10/3)/c.c**8 )
          return dist
```

```
[14]: def chirp_mass_from_chirp_correct(f_dot, f_gw):
          return 1/c.G * ((5*np.pi/96) *c.c**5 * f_dot * (np.pi * f_gw)**(-11/3))**(3/
       ↪5)
```

```
[15]: fdot = 1.1e-16 * u.Hz / u.s
      f_gw = 2.75 * u.mHz
      f_orb = f_gw / 2
      strain = 1.8e-22
```

```
[16]: chirp_mass_from_class = chirp_mass_from_chirp_from_class(fdot, f_gw)
      print(chirp_mass_from_class.to(u.Msun))
      print(chirp_mass_from_class.to(u.kg))
```

```
0.3177925430079152 solMass
6.3190182935117344e+29 kg
```

```
[17]: chirp_mass_correct = chirp_mass_from_chirp_correct(fdot, f_gw)
      print(chirp_mass_correct.to(u.Msun))
      print(chirp_mass_correct.to(u.kg))
```

```
0.6315876451010213 solMass
1.2558551077298081e+30 kg
```

```
[18]: distance_from_class = distance_from_strain_chirp_mass(strain,
       ↪chirp_mass_from_class, f_gw)
      print(distance_from_class.to(u.kpc))
```

```
2.4263943841699924 kpc
```

```
[19]: distance_correct = distance_from_strain_chirp_mass(strain, chirp_mass_correct,
       ↪f_gw)
      print(distance_correct.to(u.kpc))
```

```
7.622742772019979 kpc
```

Based on the chirp and the observed frequency, the chirp mass suggests that the source is a binary white dwarf!

2 pts off for incorrect chirp mass

2 pts off for incorrect distance

2 pts off for incorrect interpretation